INVERSE LIMITS OF RINGS ALONG SURJECTIONS WITH LOCALLY NILPOTENT KERNELS ARE HENSELIAN PAIRS

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Allowed standard prerequisites (editorial).

- An element that is a unit modulo a locally nilpotent kernel is a unit. Algebra, Lemma \ref{algebra-lemma-locally-nilpotent-unit}, cited at proof lines 2460-2461. Shows that elements which are units at some stage are units in the limit.
- Elements of one plus an ideal consisting of such units force the ideal into the Jacobson radical. Algebra, Lemma \ref{algebra-lemma-contained-in-radical}, cited at proof lines 2465-2466. Puts the kernel ideal inside the Jacobson radical, one half of the henselian condition.
- Lifting of factorisations of monic polynomials through surjections with locally nilpotent kernels. Lemma \ref{lemma-locally-nilpotent-henselian}, cited at proof lines 2471-2472. Provides the successive lifting of the factorisation of the reduction of the monic polynomial.

Source reference. Original lemma 0CT7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16057–16057. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0AMG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13404–13404. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0AME: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13402–13402. Navigation pointer to a statement the

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Source reference. Original lemma 0ALI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13371–13371. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $A = \lim A_n$ where (A_n) is an inverse system of rings whose transition maps are surjective and have locally nilpotent kernels. Then (A, I_n) is a henselian pair, where $I_n = \mathrm{Ker}(A \to A_n)$.*

Proof. Fix n. Let $a \in A$ be an element which maps to 1 in A_n. By Algebra, Lemma 0AMG we see that a maps to a unit in A_m for all $m \geq n$. Hence a is a unit in A. Thus by Algebra, Lemma 0AME the ideal I_n is contained in the Jacobson radical of A. Let $f \in A[T]$ be a monic polynomial and let $\overline{f} = g_n h_n$ be a factorization of $\overline{f} = f \bmod I_n$ with $g_n, h_n \in A_n[T]$ monic generating the unit ideal in $A_n[T]$. By Lemma 0ALI we can successively lift this factorization to $f \bmod I_m = g_m h_m$ with g_m, h_m monic in $A_m[T]$ for all $m \geq n$. At each step we have to verify that our lifts g_m, h_m generate the unit ideal in $A_n[T]$; this follows from the corresponding fact for g_n, h_n and the fact that $\mathrm{Spec}(A_n[T]) = \mathrm{Spec}(A_m[T])$ because the kernel of $A_m \to A_n$ is locally nilpotent. As $A = \lim A_m$ this finishes the proof. □